Lecture 2. Data Visualization

Why visualise, taxonomy of data graphics, a directory of plots, distributions, associations, trends, telling a story

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These notes follow the course slides (based on *Modern Data Science with R*, Baumer, Kaplan and Horton, and Wilke's *Fundamentals of Data Visualization*). All R code was run when the notes were compiled; lines starting with `##` are output. Data sets come from R packages so that every example can be reproduced; where the slides used other data, the notes say so. Material marked [Supplement] is not on the slides.

2.1 Why visualise data

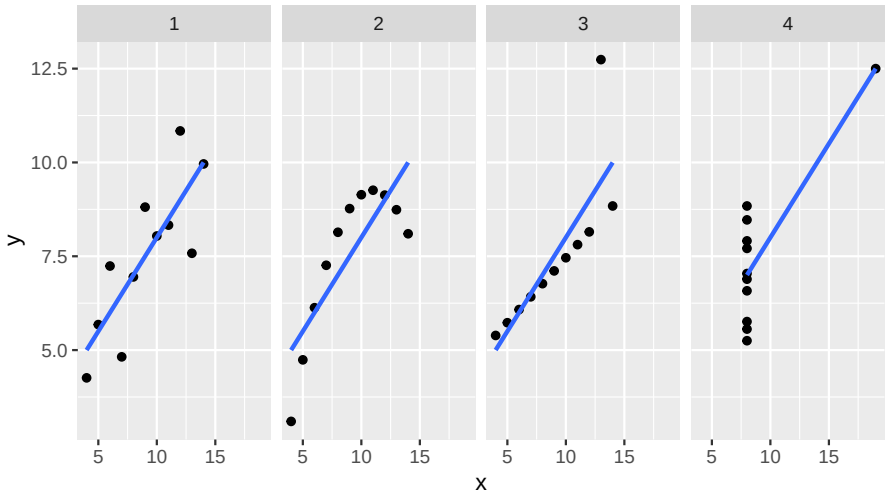
Motivation. Summary statistics compress a data set into a few numbers, and very different data sets can share the same numbers. A plot shows the structure that the numbers hide, which is why visualisation is part of analysis rather than decoration.

Visualisation lets us explore data and find patterns that would be missed if we relied only on numerical summaries. The classic illustration is **Anscombe's quartet**: four data sets with (almost) identical means, variances, correlations and regression lines.

```
anscombe_long <- anscombe %>%
  pivot_longer(everything(), names_to = c(".value", "set"), names_pattern = "(.)(.)")
anscombe_long %>%
  group_by(set) %>%
  summarise(mean_x = mean(x), mean_y = mean(y), sd_y = sd(y), cor = cor(x, y))
```

```
## # A tibble: 4 x 5
##   set   mean_x mean_y  sd_y   cor
##   <chr>  <dbl>  <dbl> <dbl> <dbl>
## 1 1          9   7.50  2.03 0.816
## 2 2          9   7.50  2.03 0.816
## 3 3          9   7.5   2.03 0.816
## 4 4          9   7.50  2.03 0.817
```

```
ggplot(anscombe_long, aes(x, y)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE, formula = y ~ x) +
  facet_wrap(~ set, nrow = 1)
```



Only the first set looks like the textbook picture of a linear relationship: the second is a curve, the third has one outlier, and the fourth is determined by a single point. Visualisation "forces us to notice what we never expected to see" (Tukey, 1977).

There is no fixed procedure. Any data set can be visualised in many ways, some informative and some not; we should look at the data from different perspectives and at coarser and finer levels,

starting simply and refining by trial and error. Tufte's guiding principles: above all show the data; avoid distorting what the data have to say; present many numbers in a small space; encourage the eye to compare; reveal the data at several levels of detail.

The main tools are histograms, density plots and bar plots (distributions of continuous and discrete variables), boxplots (distributions, tied to robust summaries), and scatterplots and their variants (bivariate data); heat maps and maps serve more specialised purposes.

2.1.1 Example: campaign spending in 2012

The slides analyse US Federal Election Commission data on spending in the 2012 elections. Three lessons from that example:

- **A third variable can reverse the story.** Total spending on each candidate made Obama look like the best-funded candidate, but colouring by type of spending showed that $251 million of the $274 million spent on his campaign was spent *against* him (92%, against 70% for Romney).
- **Faceting adds context.** Splitting spending by office (House, Senate, President) showed that only in the presidential race was more money spent attacking Democrats than Republicans.
- **Proportions, not totals.** Relating money to votes in House races, the vote share should be compared with the *share* of spending, since districts differ in population and turnout. Skewed variables (donation amounts) are better seen on a log scale, and distributions are easier to compare overlaid or stacked than side by side.

2.2 A taxonomy for data graphics

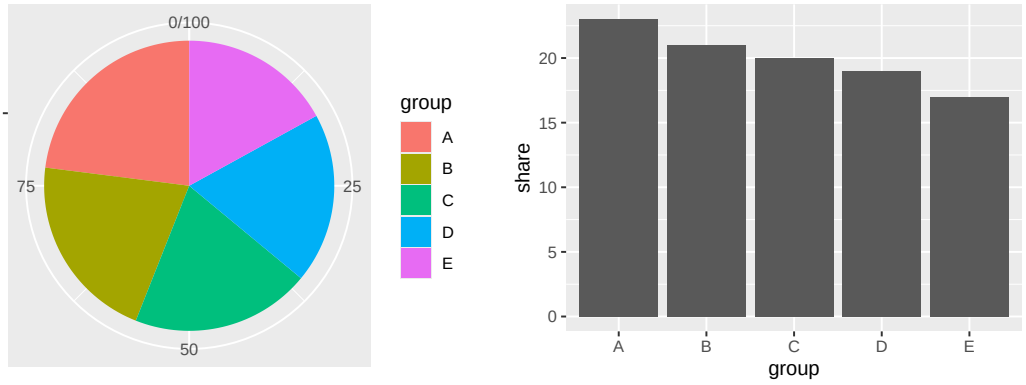
Motivation. To design a graphic deliberately, and to criticise one, it helps to break it into parts. Yau's taxonomy of four elements, which matches the grammar of graphics implemented in ggplot2, does this.

Every data graphic can be described by four elements (Yau, 2013):

1. **Visual cues**: graphical elements that draw the eye. In decreasing order of how accurately people perceive differences: position, length, angle, direction, shape (categorical), area, volume, shade and colour.
2. **Coordinate system**: Cartesian, polar, or geographic.
3. **Scale**: how distance in the graphic translates into differences in quantity. Numeric scales may be linear, logarithmic or percentage; categorical scales may be unordered or ordinal; time is numeric but has calendar units and can be periodic.
4. **Context**: titles, axis labels, reference points or lines that help the viewer make meaningful comparisons.

Consequences of the perception ranking: people judge positions and lengths well but angles and areas poorly, so **bar charts beat pie charts** and word clouds or bubble sizes convey magnitudes only roughly. Differences in colour are also hard to judge (heat maps), and red–green contrasts fail for colour-blind viewers.

```
shares <- tibble(group = LETTERS[1:5], share = c(23, 21, 20, 19, 17))
ggplot(shares, aes(x = "", y = share, fill = group)) +
  geom_col(width = 1) + coord_polar(theta = "y") + labs(x = NULL, y = NULL)
ggplot(shares, aes(group, share)) + geom_col()
```



The same five shares: in the pie the slices look equal, in the bar chart the ordering is obvious.

Dissecting a graphic. For any data graphic ask: which variables are shown and of what type; which visual cue encodes each; which coordinate system and scales are used; how context is provided. For example, a bar chart of average math SAT scores by state shows a continuous variable (score) and a categorical one (state) using length, in Cartesian coordinates with linear scales, with context from axis labels and a title.

2.3 A directory of visualisations

Motivation. Most plotting questions fall into a few types: amounts, distributions, proportions, relationships, trends and geography. Knowing the standard plot for each type is the quickest route to a good first graphic.

what to show	plots
amounts	bars (vertical or horizontal), dots where the bar would end
one distribution	histogram, density (need a binwidth or bandwidth); empirical CDF, QQ plot (faithful but harder to read)
several distributions	boxplots, violins, overlapping densities, ridgeline plots (many groups)
proportions	side-by-side bars, stacked bars (many conditions), pies (only for simple fractions of a whole); stacked densities when proportions change along a continuous variable
multiple groupings	mosaic plots, treemaps, parallel sets
x–y relationships	scatterplots, bubble charts (third variable as size), 2-d bins for many points, correlograms for many variables, line graphs when x is time
geospatial	choropleth maps, cartograms

2.4 Visualising distributions

Motivation. A histogram or density estimate depends on a tuning choice, the bin width or bandwidth, and different choices can tell different stories. Knowing the standard rules gives a principled starting point.

2.4.1 Histograms and bin width

A histogram depends on the bin width h: too narrow and noise dominates, too wide and structure disappears (the slides show Titanic passenger ages with widths of 1, 3, 5 and 15 years). Two common rules:

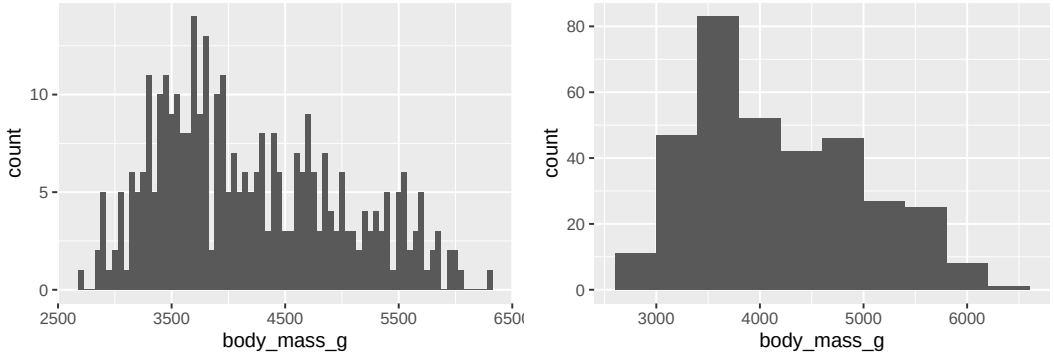
$$\text{Scott: } h = 3.49\,\hat{\sigma}\,n^{-1/3}, \qquad \text{Freedman–Diaconis: } h = 2\,\mathrm{IQR}\,n^{-1/3},$$

the second being less sensitive to outliers. For penguin body mass:

```
x <- na.omit(penguins$body_mass_g)
c(n = length(x), scott = 3.49 * sd(x) * length(x)^(-1/3),
  fd = 2 * IQR(x) * length(x)^(-1/3))
```

```
##        n    scott       fd
## 342.0000 400.2209 343.1910
```

```
ggplot(penguins, aes(body_mass_g)) + geom_histogram(binwidth = 50, na.rm = TRUE)
ggplot(penguins, aes(body_mass_g)) + geom_histogram(binwidth = 400, na.rm = TRUE)
```



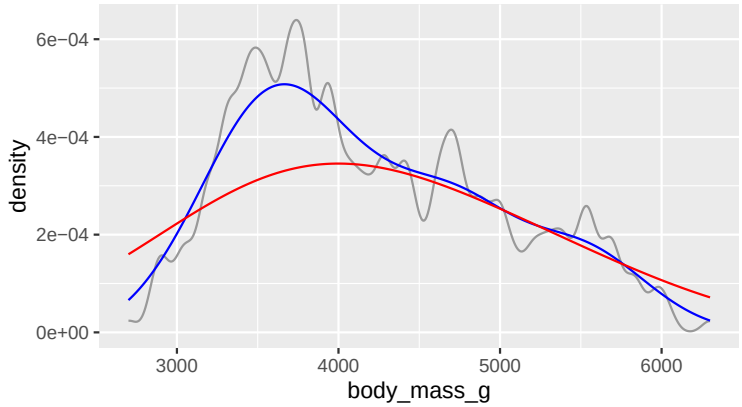
2.4.2 Density estimates

A kernel density estimate places a small bump (the **kernel** K) on each observation and adds them up:

$$\hat{f}(x) = \frac{1}{nh} \sum_{i=1}^{n} K\Big(\frac{x - X_i}{h}\Big).$$

It depends on the kernel and, much more, on the bandwidth h. A rule of thumb (Härdle, 1991) is $h = 1.06\, S\, n^{-1/5}$, where S is the standard deviation; robust versions replace S by the scaled IQR $(X_{(0.75n)} - X_{(0.25n)})/1.34$ or the scaled MAD $1.4826 \cdot \text{median}|X_i - \text{median}(X)|$.

```
ggplot(penguins, aes(body_mass_g)) +
  geom_density(bw = 50, colour = "grey60", na.rm = TRUE) +
  geom_density(bw = 1.06 * sd(x) * length(x)^(-1/5), colour = "blue", na.rm = TRUE) +
  geom_density(bw = 800, colour = "red", na.rm = TRUE)
```



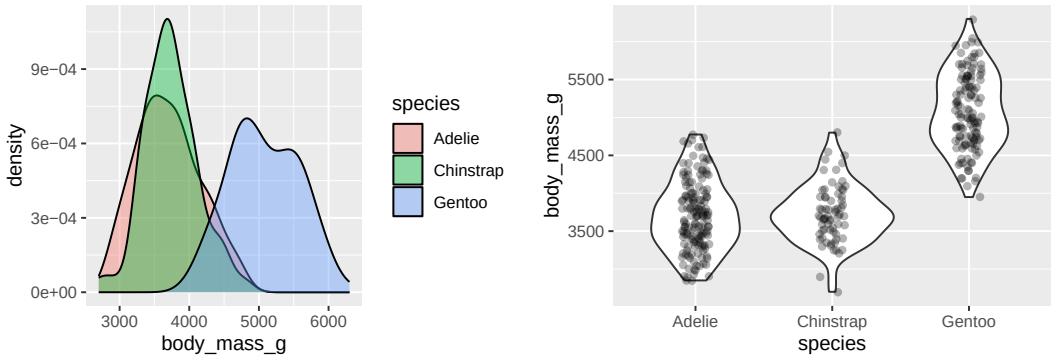
Where the 1.06 comes from [Supplement]

For a normal density with standard deviation σ and a Gaussian kernel, the bandwidth minimising the asymptotic mean integrated squared error is $h^* = (4/3)^{1/5}\sigma n^{-1/5} \approx 1.06\,\sigma n^{-1/5}$. The rule is therefore optimal for normal data and oversmooths multimodal data, as the blue curve above (which blurs the species) shows.

2.4.3 Comparing distributions

For two distributions, overlay densities or histograms (with transparency) rather than placing them side by side. For several, use jittered points, boxplots, violins (mirrored densities) or ridgeline plots (densities staggered vertically, good for many groups; package `ggridges`).

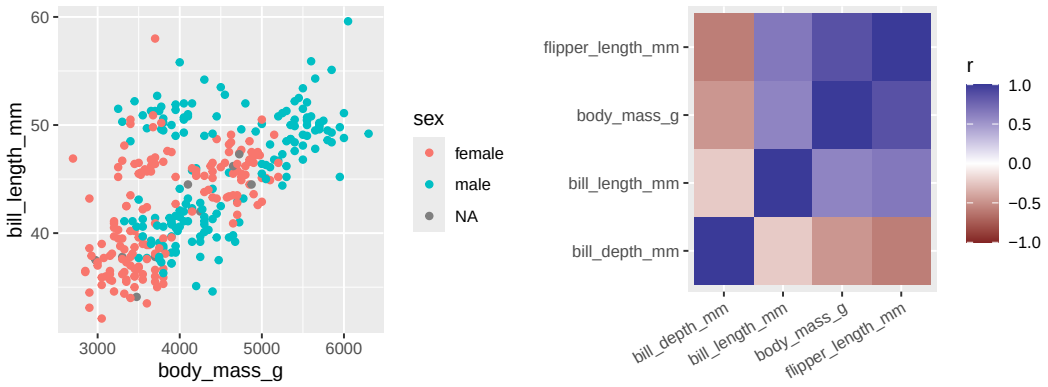
```
ggplot(penguins, aes(body_mass_g, fill = species)) +
  geom_density(alpha = 0.4, na.rm = TRUE)
ggplot(penguins, aes(species, body_mass_g)) +
  geom_violin(na.rm = TRUE) + geom_jitter(width = 0.1, alpha = 0.3, na.rm = TRUE)
```



2.5 Visualising associations

The slides use 123 blue jays with head length, skull size and body mass. With `penguins` the same steps are: a scatterplot for two variables; colour for a grouping variable; size for a third quantitative variable (a **bubble chart**, which makes associations hard to judge); a scatterplot matrix of all pairs; and a **correlogram**, a picture of the correlation matrix.

```
ggplot(penguins, aes(body_mass_g, bill_length_mm, colour = sex)) +
  geom_point(na.rm = TRUE)
penguins %>%
  select(bill_length_mm, bill_depth_mm, flipper_length_mm, body_mass_g) %>%
  cor(use = "complete.obs") %>%
  as_tibble(rownames = "x") %>%
  pivot_longer(-x, names_to = "y", values_to = "r") %>%
  ggplot(aes(x, y, fill = r)) + geom_tile() +
  scale_fill_gradient2(limits = c(-1, 1)) +
  theme(axis.text.x = element_text(angle = 30, hjust = 1)) + labs(x = NULL, y = NULL)
```



`GGally::ggpairs()` draws the full scatterplot matrix in one call.

Simpson's paradox in the penguins [Supplement]

Pooled over species, bill depth and bill length are negatively correlated; within each species the correlation is positive. A grouping variable can reverse an association, which is exactly why colour or facets for a third variable matter.

```
penguins %>% drop_na() %>%
  summarise(pooled = cor(bill_length_mm, bill_depth_mm))
```

```
## # A tibble: 1 x 1
##   pooled
##    <dbl>
## 1 -0.229
```

```
penguins %>% drop_na() %>% group_by(species) %>%
  summarise(within = cor(bill_length_mm, bill_depth_mm))
```

```
## # A tibble: 3 x 2
##   species   within
##   <fct>      <dbl>
## 1 Adelie     0.386
## 2 Chinstrap  0.654
## 3 Gentoo     0.654
```

2.6 Visualising trends

Motivation. Time series and other ordered data mix long-term trends with short-term noise. Smoothing separates them, and a transformation can turn a curved trend into a straight one that is easy to read.

Defined functional forms. The slides show monthly preprint submissions to bioRxiv growing exponentially, $y = Ae^{mx}$. Rather than fitting an exponential curve, take logs: $\log y = \log A + mx$ is linear, so fit a line to the log-transformed data.

Smoothing. To show a long-term trend while de-emphasising fluctuations (the slides use the Dow Jones index in 2009), use

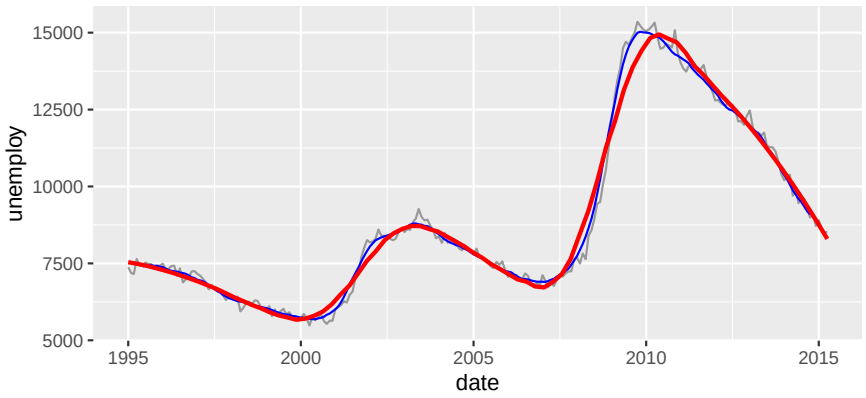
- a **moving average**: the mean over a window of neighbouring times;
- **LOESS** (locally estimated scatterplot smoothing): at each x_0, take the k nearest observations $N_k(x_0)$ and fit a line $\ell(x) = \beta_0 + \beta_1(x - x_0)$ by weighted least squares, minimising

$$\sum_{i:X_i \in N_k(x_0)} \{Y_i - \ell(X_i)\}^2 \, W\Big(\frac{|x_0 - X_i|}{\Delta(x_0)}\Big),$$

where $\Delta(x_0) = \max_{i:X_i \in N_k(x_0)} |X_i - x_0|$ and $W(u) = (1 - u^3)^3$ (tri-cube weight). The **span** k/n controls smoothness; a quadratic can replace the line;

- **splines**: piecewise polynomials joined smoothly at *knots*; faster than LOESS for large data, but different spline types and numbers of knots can give quite different curves (Lecture 10).

```
econ <- economics %>%
  filter(date >= as.Date("1995-01-01")) %>%
  mutate(ma12 = stats::filter(unemploy, rep(1/12, 12), sides = 2))
ggplot(econ, aes(date, unemploy)) +
  geom_line(colour = "grey60") +
  geom_line(aes(y = ma12), colour = "blue", na.rm = TRUE) +
  geom_smooth(method = "loess", span = 0.3, se = FALSE, colour = "red", formula = y ~ x)
```



US unemployment (thousands, `ggplot2::economics`): raw series (grey), 12-month moving average (blue) and LOESS with span 0.3 (red).

2.7 Telling a story

Motivation. A graphic made for a decision must make its point to people who did not do the analysis. The Challenger disaster is the classic case of the right data shown the wrong way.

On 27 January 1986 engineers at Morton Thiokol recommended delaying the launch of the space shuttle Challenger, fearing that the cold forecast would compromise the rubber O-rings sealing the solid rocket motors. Their 13 charts did not persuade NASA, the launch went ahead, and the O-rings failed 73 seconds after launch, killing seven astronauts. The charts showed O-ring damage for each previous flight but did not plot damage against temperature, and the hearings afterwards showed only flights with damage. Tufte's reconstruction plots damage against launch temperature for *all* flights: every launch below about 65°F had damage, and the forecast temperature (about 30°F) lay far below every previous launch, an extrapolation that by itself argued against launching.

Making a point simply. A plot of mean arrival delay for each carrier departing New York over time is rich but cluttered. If the message is "American and Delta have the shortest delays", a bar chart conveys it directly, and a second bar chart of flight counts shows that both are major NYC carriers:

```
by_carrier <- flights %>%
  group_by(carrier) %>%
  summarise(delay = mean(arr_delay, na.rm = TRUE), n = n()) %>%
  filter(n > 10000)
ggplot(by_carrier, aes(reorder(carrier, delay), delay)) + geom_col() +
  labs(x = "carrier", y = "mean arrival delay (min)")
```

```
ggplot(by_carrier, aes(reorder(carrier, delay), n)) + geom_col() +
  labs(x = "carrier", y = "number of flights")
```

